9/1/2026

N + - Network Fundamentals & Building Networks.

Weekly Review and Practice.

YAGNIK GHETIYA

TASK 1.

❖ On Cisco Packet Tracer, create a network with three switches and assign at least two different VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support'). Connect two PCs to each switch and assign each PC to the correct VLAN based on department.

- **Creating a Three-Switch VLAN Network:**

Network Setup: Three switches (Switch1, Switch2, and Switch3) were connected. Two PCs were connected to each switch.

Create VLANs: The following VLANs were configured on all three switches.

Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Support

Assign PCs to VLANs:

Switch	PC	VLAN	Department
Switch1	PC1	10	Sales
Switch1	PC2	20	Support
Switch2	PC3	10	Sales
Switch2	PC4	20	Support
Switch3	PC5	10	Sales
Switch3	PC6	20	Support

Result: Three switches were configured with Sales and Support VLANs, and each PC was assigned to its department's VLAN.

TASK 2.

❖ Configure trunk ports between the switches to allow VLAN traffic to pass through. Verify trunking by using the show interfaces trunk command and submit a screenshot of the output.

➢ **Task: Configure and Verify Trunk Ports:**

Configure Trunk Ports: On each switch, configure the interfaces connecting to other switches as trunk ports.

Switch(config)# interface fa0/1
Switch(config-if)# switchport mode trunk

Verify Trunking: Run this command on each switch:

Switch# show interfaces trunk

Result: The trunk ports allow VLAN 10 (Sales) and VLAN 20 (Support) traffic to pass between switches.

Screenshot:

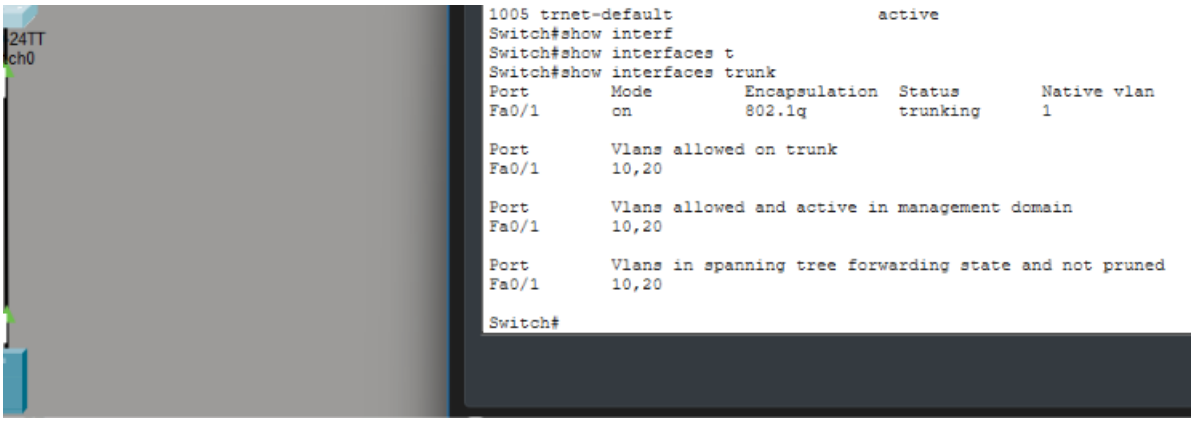


Figure : Verification of Trunk Ports Using the show interfaces trunk Command in Cisco Packet Tracer.

TASK 3.

❖ Set up a Layer 3 device (Router-on-a-Stick using a router or a multilayer switch) to enable inter-VLAN routing between VLAN 10 and VLAN 20. Assign appropriate sub-interfaces and IP addresses for each VLAN.

- **Configure Inter VLAN Routing Using Router on a Stick:**

Configure Router Sub-Interfaces

```
Router(config)# interface g0/0
Router(config-if)# no shutdown
Router(config)# interface g0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config)# interface g0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

Configure the Switch Trunk Port

```
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode trunk
```

Result: Router sub-interfaces were configured for VLAN 10 and VLAN 20, enabling inter-VLAN routing between the two networks.

TASK 4.

❖ Test connectivity by pinging between a PC in VLAN 10 and a PC in VLAN 20. If pings fail, troubleshoot and fix the configuration so that inter-VLAN communication works, Check sub-interface IPs, trunk configuration, and default gateways on PCs.

- **Testing and Troubleshooting Inter-VLAN Connectivity:**

Connectivity Test: A ping test was performed from a PC in VLAN 10 to a PC in VLAN 20.

ping 192.168.20.2

Troubleshooting: The router sub-interface IP addresses, switch trunk configuration, and PC default gateways were checked and corrected where necessary.

VLAN	PC IP Address	Default Gateway
VLAN 10	192.168.10.2	192.168.10.1
VLAN 20	192.168.20.2	192.168.20.1

Result: Successful ping replies confirmed that inter-VLAN communication was working correctly.